

.Net + Angular Full Stack Development

Become a Front-End / Full-Stack Developer



Course Curriculum

Introduction to Full-Stack Development

- Overview of Front-End, Back-End, and Database
- Role of Angular, C#, .NET Core, and SQL Server in Full-Stack Development
- Setting Up the Development Environment: Visual Studio, SQL Server, Angular CLI, and Node.js

Front-End Development

Module-1 : HTML, CSS, and Bootstrap

- HTML Fundamentals: Structure of an HTML Document, Semantic HTML5, Forms, Tables, and Multimedia
- CSS Basics: Selectors, Box Model, Flexbox, Grid Layout, Styling Forms, Animations, and Transitions
- Bootstrap: Grid System and Breakpoints, Components (Navbar, Cards, Buttons, Forms), Customizing Themes
- Responsive Design: Using Media Queries, Mobile-First Design Principles

Module-2 : Angular Framework

- Introduction to Angular: SPA (Single Page Application) Overview, Setting up an Angular Project, Angular Architecture (Modules, Components, Templates)
- Forms in Angular: Template-Driven and Reactive Forms, Validation and Error Handling
- Angular Material: Material Design Components, Navigation, Dialogs, Tables, and Theme Customization
- Routing and Navigation: Setting up Routes and Lazy Loading, Router Guards
- CRUD Operations: Consuming REST APIs, Creating a CRUD Application with Angular and APIs

Module-3 : C# Programming

- Basics: Variables, Data Types, and Operators, Control Structures (If, Switch, Loops), Methods and Functions
- Object-Oriented Programming: Classes, Objects, and Constructors, Inheritance, Polymorphism, Abstraction, Interfaces
- Advanced C#: Delegates, Events, and Lambda Expressions, Exception Handling, LINQ (Language Integrated Query)

Module-4 : .NET Core 8

- Introduction to .NET Core: Overview of .NET Core 8, Setting up a Web API Project in .NET Core
- RESTful APIs: Creating Controllers and Routes, Handling HTTP Methods (GET, POST, PUT, DELETE), Dependency Injection in .NET Core
- Middleware in .NET Core: Custom Middleware, Authentication and Authorization (JWT)
- Working with Configuration and Services
- Error Handling and Logging

Module-5 : Entity Framework Core

- Introduction to Entity Framework Core: Code-First and Database-First Approaches, Setting up DbContext
- Entity Relationships: One-to-One, One-to-Many, and Many-to-Many, Navigational Properties
- Migrations: Creating and Applying Migrations, Updating the Database Schema
- CRUD Operations with Entity Framework Core: Adding, Updating, Deleting, and Retrieving Data

Module-6 : SQL Server

- Introduction to SQL Server: Installing and Configuring SQL Server, SQL Server Management Studio (SSMS)
- SQL Basics: DDL, DML, and DQL Statements, Creating and Managing Tables, Writing Queries (SELECT, INSERT, UPDATE, DELETE)
- Advanced SQL: Joins (INNER, OUTER, LEFT, RIGHT), Subqueries and Common Table Expressions (CTEs), Stored Procedures and Functions, Triggers and Indexing
- Database Design: Normalization and Denormalization, Designing Relational Databases, Creating Relationships with Primary and Foreign Keys

Module-7 : Full-Stack Integration

- Building a Complete Full-Stack Application:Example (E-Commerce or Task Management System)
- Integration: Using Angular to Consume .NET Core APIs, Persisting Data in SQL Server via Entity Framework
- Authentication and Authorization: JWT-Based Authentication in .NET Core, Protecting Routes in Angular
- Real-Time Updates: SignalR Integration (optional)

Module-8 : Deployment

- Deploying the Application: Hosting Angular on Firebase/Netlify, Hosting .NET Core APIs on Azure or IIS
- Database Management: Backing up and Restoring Databases in SQL Server

Module-9 : Capstone Project

- Develop a Full-Stack Application integrating Angular,.NET Core, Entity Framework, and SQL Server.
- Example Projects: Online Store, Employee Management System, Learning Management System

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